Can AI writing be salvaged? Mitigating Idiosyncrasies and Improving Human-AI Alignment in the Writing Process through Edits

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Abstract

LLM-based applications are helping people write, and LLM-generated text is making its way into social media, journalism, and our classrooms. However, the differences between LLM-generated and human-written text remain unclear. To explore this, we hired professional writers to edit paragraphs in several creative domains. We first found these writers agree on undesirable idiosyncrasies in LLM-generated text, formalizing it into a seven-category taxonomy (e.g. clichés, unnecessary exposition). Second, we curated the LAMP corpus: 1,057 LLM-generated paragraphs edited by professional writers according to our taxonomy. Analysis of LAMP reveals that none of the LLMs used in our study (GPT4o, Claude-3.5-Sonnet, Llama-3.1-70b) outperform each other in terms of writing quality, revealing common limitations across model families. Third, building on existing work in automatic editing we evaluated methods to improve LLM-generated text. A large-scale preference annotation confirms that although experts largely prefer text edited by other experts, automatic editing methods show promise in improving alignment between LLM-generated and human-written text.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Empirical studies in collaborative and social computing**; • **Computing methodologies** → *Natural language generation.*

Keywords

Human-AI collaboration, Large Language Models, Design Methods, Text Editing, Natural Language Generation, Evaluation, Writing Assistance, Generative AI, Homogenization, Alignment, Behavioral Science

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1 Introduction

Artificial Intelligence (AI) has the potential to revolutionize how we write, communicate, and express ideas [68]. Recent studies have demonstrated the potential of large language models (LLMs) in assisting with various writing tasks, including argumentative [24, 65], scientific [43], and creative writing [16, 51, 77, 78, 117]. Aligning LLMs with human preferences [86] has enabled their transformation into user-friendly tools for non-technical users, such as Google's WorkSpace Labs, Grammarly, and Sudowrite. However, to truly benefit society, AI writing assistants must enhance human creativity and expression rather than homogenize content or diminish linguistic diversity [39, 52]

While LLM-based writing assistants have the potential to improve writing quality and increase author productivity, they also introduce an algorithmic monoculture [58]. Padmakumar and He [87] and Anderson et al. [3] discuss how writing with LLMs unintentionally reduces content diversity, leading to homogenization. This homogenization occurs not just at the semantic level but also at syntactic (structural), lexical [59], and stylistic levels [103]. For instance, prior work from Chakrabarty et al. [16], Ippolito et al. [51] has shown how LLM-generated text is often hackneyed and rife with clichés while failing to demonstrate rhetorical complexity and often revealing the subtext —a phenomenon known as "telling instead of showing" [15]. Additionally, LLM-generated texts are typically full of redundant exposition, overwrought metaphors, and florid descriptions due to verbosity bias during preference labeling [101].

Current AI-assisted writing tools are powered by pre-trained language models that are refined through reinforcement learning from human feedback (RLHF) [121]. RLHF transforms human preferences into training data to guide language models toward desired outcomes. The most common type of feedback used with RLHF is binary preferences between pairs of examples sampled from one or more language models [19] (See Fig 1 Traditional Alignment). However, this approach has a drawback. The paired outputs may differ in numerous ways and could be equally flawed in containing idiosyncrasies. Asking an annotator to choose between two undesirable outputs does not improve alignment[1]. [14, 44]. We argue that alignment training needs to be aware of how desirable any individual response is, regardless of its preference relationship. Editing undesirable portions of a response can be seen as an effective mechanism for enhancing alignment (See Fig 1 Alignment via Edits). An LLM-generated response that has been edited typically contains

[1]In the current design of RLHF annotators are not allowed to not pick either

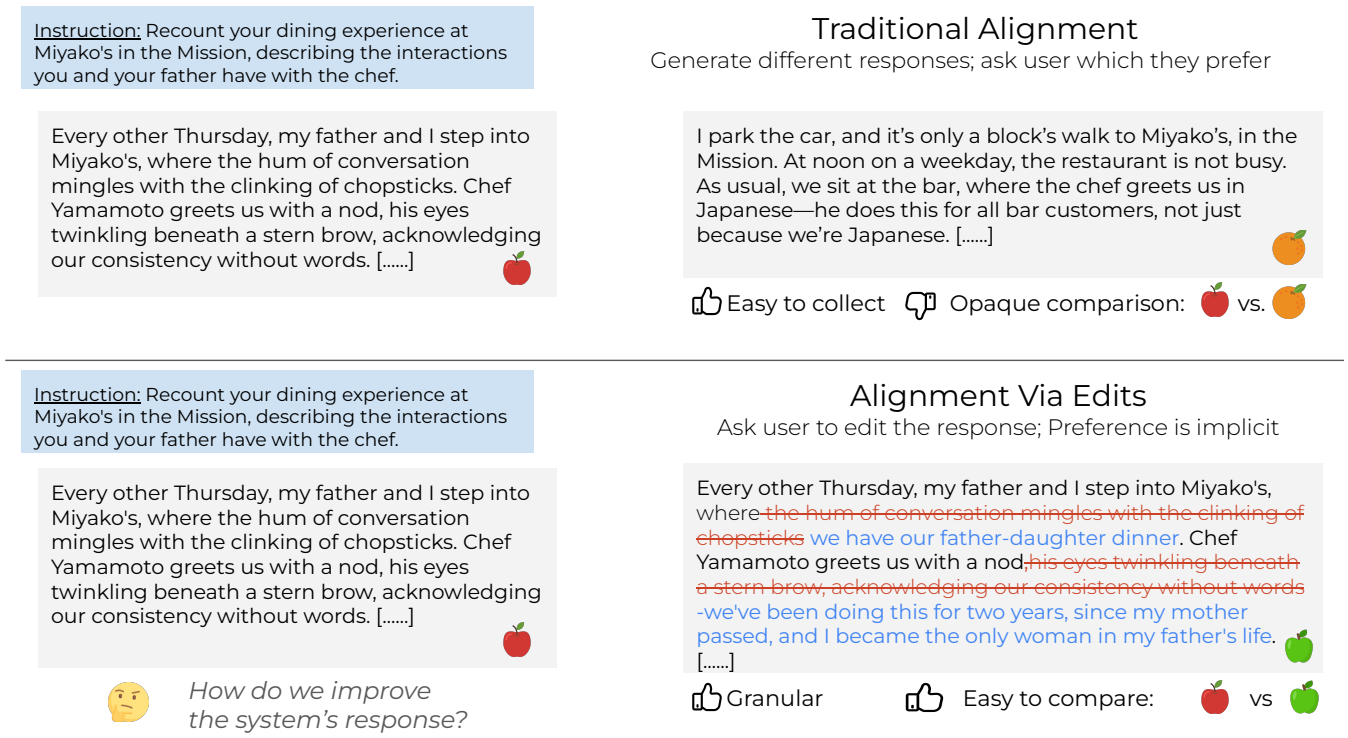

Figure 1: To align models to human preferences, human annotators are typically shown two responses and asked to choose the one they prefer. (i) The top portion of the Figure shows Traditional Alignment: it is often hard to compare two responses that differ widely. (ii) The bottom portion of the Figure shows Alignment via Edits where the original response is edited, allowing for a more granular comparison, with the edited version of the text naturally preferred over the original response.

fewer undesirable traits and can be paired with the original LLM-generated response for preference ranking (*edited* > *original*). The challenge, however, lies in consistently identifying and implementing edits that enhance quality while aligning with human preferences for effective writing. Prompting techniques that encourage the model to self-edit [72] have shown promise; however, they do not work for long-form or paragraph-level writing [88]. The primary reason for this is that LLMs do not inherently know what aspects of the writing need improvement, how extensively they should edit, or how to make changes that align with any given writer's expertise level and style.

To address these problems, we first create a comprehensive taxonomy of edit categories based on expert writing practices. We then recruit 18 writers to edit LLM-generated text using categories from our taxonomy. We define edits as changes that alter, replace, or refine specific phrases, clauses, or sentences within a larger text. We restrict our focus to generating text in literary fiction and creative non-fiction, as these genres challenge LLMs with their creativity, emotional nuance, and sophisticated language use. We focus on paragraph-level edits, as they balance granularity and scope, reducing costs and annotator fatigue. Paragraphs capture style and context better than sentences, enabling more cohesive improvements. Given LLMs' limitations in long-term discourse coherence, paragraph-level enhancements facilitate human-AI collaboration. This approach allows humans to guide the overall structure and flow, while AI handles the lower-level details [24]. Finally, while expert writer's edits are valuable for identifying idiosyncrasies, this approach may not scale for large text volumes. To address this, we design few-shot prompts that use writer's edits to identify problematic spans in LLM-generated paragraphs and suggest improvements. This method aims to enhance overall paragraph quality at scale. Our work adds to the existing rich literature [40, 53, 70, 116] on using completion edits to improve alignment in different tasks. To summarize our contributions:

- We propose a comprehensive edit taxonomy informed by expert writing practices that serve as a useful framework to identify and mitigate the idiosyncrasies in LLM-generated text.
- We release the LAMP (Language model Authored, Manually Polished) corpus containing 1057 <instructions, response> pairs grounded in real-world writing tasks such as Literary Fiction or Creative Non-Fiction. These responses originally generated by LLMs are further edited by 18 professional writers using the above-mentioned taxonomy, resulting in 8035 fine-grained edits (Section 4).
- We present a thorough analysis of the process of editing LLM-generated text, offering insights into how expert writers edit them, to what extent the edits differ in quantity, how the

distribution of edit categories varies across text generated by different model families, and whether LLM generated text contain any specific stylistic idiosyncrasies (Sections 4 and 5.3).
- Building on prior work Hayes et al. [47], Scardamalia [102] we conduct an empirical investigation that tests if LLMs can automatically detect and rewrite their own idiosyncrasies. Our statistically significant results show an encouraging preference trend $Writer - edited > LLM - edited > LLM - generated$ suggesting that edits improve human-AI alignment in the writing process.
- Finally, we discuss how LLM edits can both mimic and differ from edits provided by professional writers, and what future LLM-based writing support tools can do to improve the co-writing experience.

Our code, data, and experimental setup is available at [2].

2 RELATED WORK

2.1 Text Editing in HCI

Text editing is the process of modifying written content using specialized software. HCI research on text editing aims to improve digital writing tools' efficiency and usability. Word processors have long allowed flexible editing functions [92]. Systems like Soylent [7], MicroWriter [110], and WearWrite [80] developed interfaces for crowd-based editing, focusing on task breakdown, cost management, and minimizing delays. Robertson and Black [99] proposes a goal-fate analysis model for text editing behavior, supported by data showing distinct plan units in editing tasks, with the potential for intention-based user assistance. Tyler et al. [113] investigated text editing skill progression and effective training methods. Rosson [100] examined the impact of experience on editing behavior, questioning if users naturally develop optimal strategies or plateau, noting that experienced users tend to develop more efficient editing heuristics than novices. Reza et al. [98] present ABScribe, a novel interface that streamlines the process of generating and comparing writing variations using Large Language Models, addressing challenges in existing text-editing workflows and improving writer's efficiency and satisfaction. Zhou and Sterman [119] suggest that imperfect AI text suggestions can promote deeper engagement in rewriting, potentially preserving the writer's authenticity and creative ownership. Park and Lee [91] found that providing rationales for edits in collaborative writing was generally preferred by participants, despite no significant differences in survey results. This led to design recommendations for effective collaboration. Dang et al. [21] propose a text editor with continuously updated paragraph-wise summaries as margin annotations to help users plan, structure, and reflect on their writing process.Laban et al. [61] introduce InkSync, an LLM-based editing interface suggesting executable document edits. It uses a three-stage approach (Warn, Verify, Audit) to reduce factual errors and enhance editing accuracy, efficiency, and user experience compared to standard chat interfaces. In contrast to existing research, we focus on edits as a method to improve human-AI alignment in writing assistance. Our work characterizes the undesirable aspects of AI writing informed by expert consensus and designs an approach to mitigate these through text editing.

2.2 Text Editing in NLP

NLP research has explored various text editing tasks [18, 62, 93]. Adding to it the advent of Large Language Models has enabled AI-assisted writing tools [12, 50]. Faltings et al. [34] release the WikiDocEdits dataset and propose an interactive text generation setting in which a user interacts with the system by issuing commands to edit existing text. Raheja et al. [95] proposed an instruction-based editing system using fine-tuned language models. Shu et al. [104] developed strategies for cross-sentence rewriting and introduced the OpenRewriteEval benchmark. Reid and Neubig [97] modeled multi-step editing processes to better mimic human content creation and improve performance on various tasks. Kim et al. [55] presented a system that iteratively improves fluency, clarity, coherence, and style by detecting editable spans and their corresponding edit intents, then instructing a revision model to refine the text. Yang et al. [115] developed a taxonomy and classifier for Wikipedia edit intentions. Following them Du et al. [29] created a multi-domain corpus of revised text with annotated edit intentions. Unlike existing work, we create a resource for text editing that caters to challenging writing tasks (literary fiction and creative nonfiction). Our data consists of 8035 fine-grained edits that are annotated by creative writing experts and we further show how recent advances in few-shot learning can help models improve their own writing by learning from edits provided by the writers.

2.3 Issues in AI Writing

Prior work has highlighted several issues in AI-generated text. Chakrabarty et al. [15, 16], Ippolito et al. [51], Marco et al. [74], Mirowski et al. [78] show how LLM-generated text is often hackneyed and rife with clichés, lacks nuance, subtext, and rhetorical complexity. Recent work from Mirowski et al. [77] shows LLMs fail to act as good creativity support tools for comedy writing and mostly resort to producing bland and biased comedy tropes. They further highlight how existing moderation strategies used in safety filtering and instruction-tuned LLMs reinforce hegemonic viewpoints by erasing minority groups and their perspectives in writing. In summarizing short stories Subbiah et al. [108] demonstrate how LLMs struggle with specificity and interpretation of difficult subtext. In a similar vein, Tian et al. [112] shed light on how LLM-generated stories are homogeneously positive and lack tension. Compared to existing work we create a fine-grained taxonomy highlighting the issues in AI writing and further create a large-scale corpus to fuel research in this direction. We also develop automated methods to identify and mitigate issues in AI writing at scale.

2.4 Human AI alignment in Writing

Lee et al. [64] highlight how AI tools have transformed writing processes, establishing new criteria for future AI writing assistants. In a similar vein Li et al. [68] reveal that while users benefit from AI assistance in productivity and confidence, potential drawbacks include decreased accountability and diversity in writing. LLMs used in writing assistance can significantly influence human-authored content. Hohenstein and Jung [49] found LLM-generated

[2]https://github.com/salesforce/creativity_eval/tree/main/Writing_Alignment